

UNIT-III VECTOR INTEGRATION

Topic Learning Objectives:

Upon Completion of this unit, students will be able to:

- Understand the fundamentals of the integration of vector point function.
- Solve line, surface and volume integrals.
- Apply Green's Theorem, Stokes' Theorem and Gauss' Theorem in solving engineering problems.
- Estimate and apply the concepts of solenoidal and irrotational fields to calculate integrals of vector functions.

Line Integral: Any integral which is to be evaluated along a curve is called line integral.

If $\vec{F}(x, y, z)$ is a vector point function and C is any curve then $\int_C \vec{F} \cdot d\vec{r}$ is called the vector line integral. (Tangential line integral or line integral)

NOTE:

1. C is called path of integration.
2. If $\vec{F} = f_1\hat{i} + f_2\hat{j} + f_3\hat{k}$ then $\int_C \vec{F} \cdot d\vec{r} = \int_C f_1 dx + \int_C f_2 dy + \int_C f_3 dz$
3. When C is a simple closed curve, line integral is denoted by $\oint_C \vec{F} \cdot d\vec{r}$ (means the line integral of $\vec{F}$ taken once around C in the anticlock wise direction).
4. If $\vec{F}$ represents force acting on a particle then the line integral $\int_C \vec{F} \cdot d\vec{r}$ represents work done by a force $\vec{F}$.
5. If $\vec{F}$ represents the velocity of a fluid then $\int_C \vec{F} \cdot d\vec{r}$ represents circulation of $\vec{F}$ around C .
6. Condition for $\vec{F}$ to be conservative is $\nabla \times \vec{F} = 0$.
7. If $\text{curl } \vec{F} = 0$, then $\int_C \vec{F} \cdot d\vec{r}$ is independent of path.

Line integrals compute the accumulated effect of a vector field along a path. In computer science, they are used in robotics, path planning, and physics-based simulations to calculate quantities such as work done by forces along a trajectory. In computer graphics and game engines, line integrals help model motion along curves and evaluate energy or force effects when objects move through simulated environments such as wind or magnetic fields.

Problem 1. If $\vec{F} = (5xy - 6x^2)\hat{i} + (2y - 4x)\hat{j}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ along $y = x^3$ in XY -plane from $(1, 1)$ to $(2, 8)$.

Solution: Given

$$\vec{F} = (5xy - 6x^2)\hat{i} - (2y - 4x)\hat{j}$$

$$\vec{r} = x\hat{i} + y\hat{j} \Rightarrow d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$y = x^3 \Rightarrow dy = 3x^2 dx \text{ and } x: 1 \text{ to } 2$$

Consider

$$\vec{F} \cdot d\vec{r} = (5x^3 - 6x^2)dx + (2x^3 - 4x)3x^2dx$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_1^2 (5x^4 - 6x^2 + 6x^5 - 12x^3)dx = [x^5 - 2x^3 + x^6 - 3x^4]_1^2 = 35.$$

Problem 2. Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along

- The straight line $(0, 0, 0)$ to $(2, 1, 3)$.
- The curve $x = 2t^2, y = t, z = 4t^2 - t$ from $t = 0$ to $t = 1$.
- The curve defined by $x^2 = 4y, 3x^3 = 8z$ from $x = 0$ to 2 .

Solution: Work done $= \int_C \vec{F} \cdot d\vec{r} = \int_C (3x^2dx + (2xz - y)dy + zdz)$ ----- (i)

a) C is a straight line joining $(0, 0, 0)$ and $(2, 1, 3)$.

The equation of the line is given by $\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$

We have $x = 2t \Rightarrow dx = 2dt, y = t \Rightarrow dy = dt, z = 3t \Rightarrow dz = 3dt$
and $t = 0$ to 1 [$\because t = y, y = 0$ to 1]

then equation (i)

$$\Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_0^1 \{(3(2t)^2)(2dt) + (2(2t)(3t) - t)dt + (3t)3dt\}$$

$$= \int_0^1 (36t^2 + 8t) dt = \left[36 \frac{t^3}{3} + 8 \frac{t^2}{2} \right]_0^1 = 16.$$

b) Given curve $x = 2t^2 \Rightarrow dx = 4t dt, y = t \Rightarrow dy = dt, z = 4t^2 - t \Rightarrow dz = (8t - 1)dt$

then (i) becomes

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \{(3(2t^2)^2)(4t dt) + (2(2t^2)(4t^2 - t) - t)dt + (4t^2 - t)(8t - 1)dt\}$$

$$= \int_0^1 (48t^5 + 16t^4 + 28t^3 - 12t^2) dt = \frac{71}{5}$$

c) Given curve $x^2 = 4y \Rightarrow y = \frac{x^2}{4} \Rightarrow dy = \frac{x}{2}dx, 3x^3 = 8z \Rightarrow z = \frac{3x^3}{8} \Rightarrow dz = \frac{9}{8}x^2dx$ and $x: 0$ to 2 , then (i) becomes

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \left\{ 3x^2dx + \left(2x \left(\frac{3}{8}x^3 \right) - \frac{x^2}{4} \right) \frac{x}{2}dx + \frac{3}{8}x^3 \frac{9}{8}x^2dx \right\}$$

$$= \int_0^2 \left(3x^2 + \frac{3}{8}x^5 - \frac{x^3}{8} + \frac{27}{64}x^5 \right) dx = 16.$$

Exercise:

- If $\vec{F} = x^2\hat{i} + xy\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, 1)$

- (a) along the line $y = x$ Ans: $\frac{2}{3}$
 (b) along the parabola $y = \sqrt{x}$ Ans: $\frac{7}{12}$.
2. Find the total work done by the force represented by $\vec{F} = 3xy\hat{i} - y\hat{j} + 2zx\hat{k}$ in moving a particle round the circle $x^2 + y^2 = 4, x = 2 \cos \theta, y = 2 \sin \theta$ & $z = 0, 0 \leq \theta \leq 2\pi$.
3. Find the circulation of $\vec{F}$ around the curve C , where C is the rectangle whose vertices are given by $(0, 0), (1, 0), (1, \frac{\pi}{2})$ & $(0, \frac{\pi}{2})$ and $\vec{F} = e^x \sin y \hat{i} + e^x \cos y \hat{j}$.
4. If $\vec{F} = (2x + y^2)\hat{i} + (3y - 4x)\hat{j}$ evaluate $\oint_C \vec{F} \cdot d\vec{r}$ around a triangle ABC in the xy -plane with $A(0, 0)$ $B(2, 0)$ and $C(2, 1)$,
 a) In the counter clockwise direction. Ans: $-\frac{14}{3}$
 b) What is the value in the opposite direction? Ans: $\frac{14}{3}$
5. Evaluate the line integral $\int_C ((x^2 + xy)dx + (x^2 + y^2)dy)$, where C : square: $x = \pm 1, y = \pm 1$. Ans: 0

NOTE: If circulation is "0" (that is $\int \vec{F} \cdot d\vec{r} = 0$), then $\vec{F}$ is irrotational.

GREEN'S THEOREM

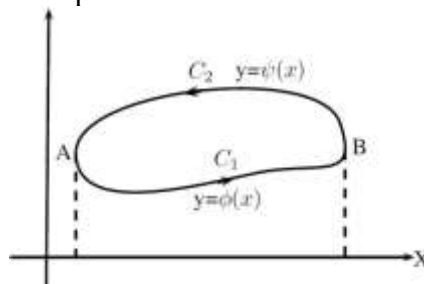
Green's theorem in the plane relates a line integral around a closed curve to a double integral over the region enclosed by the curve. In computer science, it is useful in computational geometry, image processing, and fluid simulation, where properties of a region can be determined from its boundary. It is also applied in algorithms for area computation, boundary detection, and contour analysis in digital images.

Theorem:

If R is a closed region in XY -plane, bounded by a simply closed curve C and if $P(x, y)$ and $Q(x, y), \frac{\partial}{\partial x} Q(x, y), \frac{\partial}{\partial y} P(x, y)$ be continuous functions at every point in R , then

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Proof: Suppose that C is a simply closed curve with the property that any line parallel to either axis meets the curve in at most two points.



Consider

$$\iint_R \left(-\frac{\partial P}{\partial y} \right) dx dy = \int_{x=a}^b \int_{y=\phi(x)}^{\psi(x)} \left(-\frac{\partial P}{\partial y} \right) dy dx = \int_{x=a}^b -P(x, y) \Big|_{\phi(x)}^{\psi(x)} dx$$

$$\begin{aligned}
 &= \int_{x=a}^b [-P(x, \psi(x)) + P(x, \phi(x))] dx \\
 &= \int_b^a P(x, \psi(x)) dx + \int_a^b P(x, \phi(x)) dx \\
 \Rightarrow \int_{c_1} P(x, y) dx + \int_{c_2} P(x, y) dx &= \int_c P(x, y) dx
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \iint_R \left(\frac{\partial Q}{\partial x} \right) dx dy &= \int_c Q(x, y) dx \\
 \therefore \int_c P(x, y) dx + Q(x, y) dy &= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.
 \end{aligned}$$

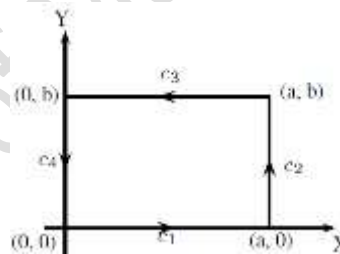
Problem 1. Verify Green's theorem in the plane for $\oint_C \{(x^2 + y)dx - xy^2 dy\}$ taken around the boundary of the rectangle whose vertices are $(0, 0)$, $(a, 0)$, (a, b) and $(0, b)$.

Solution: We have to verify

$$\oint_C P(x, y) dx + Q(x, y) dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Consider

$$\oint_C P dx + Q dy = \int_{c_1} P dx + Q dy + \int_{c_2} P dx + Q dy + \int_{c_3} P dx + Q dy + \int_{c_4} P dx + Q dy$$



$$\oint_C \{(x^2 + y)dx - xy^2 dy\} = \oint_C P dx + Q dy$$

Along C_1 : $y = 0 \Rightarrow dy = 0$ and $x: 0$ to a

$$\int_{c_1} \{(x^2 + y)dx - xy^2 dy\} = \int_0^a x^2 dx = \left. \frac{x^3}{3} \right|_0^a = \frac{a^3}{3}.$$

Along C_2 : $x = a \Rightarrow dx = 0$ and $y: 0$ to b

$$\int_{c_2} \{(x^2 + y)dx - xy^2 dy\} = \int_0^b -ay^2 dy = \left. -\frac{ay^3}{3} \right|_0^b = -\frac{ab^3}{3}.$$

Along C_3 : $y = b \Rightarrow dy = 0$ and $x: a$ to 0

$$\int_{c_3} \{(x^2 + y)dx - xy^2 dy\} = \int_a^0 (x^2 + b) dx = \left. \frac{x^3}{3} + bx \right|_a^0 = -\frac{a^3}{3} - ba.$$

Along $C_4: x = 0 \Rightarrow dx = 0$ and $y: b$ to 0

$$\int_{c_3} \{(x^2 + y)dx - xy^2 dy\} = \int_b^0 0 dy = 0$$

$$\begin{aligned} \therefore \oint_C \{(x^2 + y)dx - xy^2 dy\} &= \oint_C Pdx + Qdy \\ &= \frac{a^3}{3} - \frac{ab^3}{3} - \frac{a^3}{3} - ba + 0 = -ab \left(1 + \frac{b^2}{3}\right) \quad \dots \dots (1) \end{aligned}$$

Next consider,

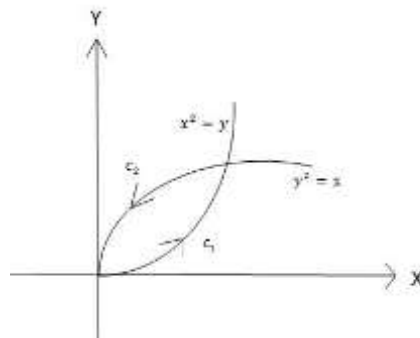
$$\begin{aligned} \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy &= \int_{x=0}^a \int_{y=0}^b (-y^2 - 1) dy dx = - \int_{x=0}^a \left[\frac{y^3}{3} + y \right]_0^b dx = - \int_0^a \left(\frac{b^2}{3} + b \right) dx \\ &= -ab \left(1 + \frac{b^2}{3}\right) \quad \dots \dots (2) \end{aligned}$$

From (1) and (2), Green's theorem is verified.

Problem 2. Verify Green's theorem in the plane for $\int_C \{(x - y)dx + (x + y)dy\}$ taken around the boundary of the finite area in the positive quadrant included between $y = x^2$ & $x = y^2$.

Solution: We have to verify

$$\begin{aligned} \int_C P(x, y)dx + Q(x, y)dy &= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \\ \int_C P dx + Q dy &= \int_{c_1} P dx + Q dy + \int_{c_2} P dx + Q dy \\ &= \int_{c_1} \{(x - y)dx + (x + y)dy\} + \int_{c_2} \{(x - y)dx + (x + y)dy\} \end{aligned}$$



Along $C_1: y = x^2 \Rightarrow dy = 2xdx$ and $x: 0$ to 1

$$\begin{aligned} \int_{c_1} \{(x - y)dx + (x + y)dy\} &= \int_0^1 \{(x - x^2)dx + (x + x^2)2xdx\} \\ &= \int_0^1 (2x^3 + x^2 + x)dx = \left[2\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} \right]_0^1 = \frac{4}{3}. \end{aligned}$$

Along $C_2: x = y^2 \Rightarrow dx = 2ydy$ and $y: 1$ to 0

$$\begin{aligned}\int_{C_1} \{(x-y)dx + (x+y)dy\} &= \int_1^0 \{(y^2-y)2ydy + (y^2+y)dy\} \\ &= \int_1^0 (2y^3 - y^2 + y)dy = \left[2\frac{y^4}{4} - \frac{y^3}{3} + \frac{y^2}{2} \right]_1^0 = -\frac{2}{3}.\end{aligned}$$

$$\therefore \int_C P dx + Q dy = \frac{4}{3} - \frac{2}{3} = \frac{2}{3} \quad \dots\dots\dots(1)$$

Now $P = x - y \Rightarrow \frac{\partial P}{\partial y} = -1$ and $Q = x + y \Rightarrow \frac{\partial Q}{\partial x} = 1$

$$\begin{aligned}\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy &= \int_{y=0}^1 \int_{x=y^2}^{\sqrt{y}} (1+1) dx dy = \int_{y=0}^1 \left(\int_{x=y^2}^{\sqrt{y}} 2 dx \right) dy \\ &= \int_{y=0}^1 (2x)_{y^2}^{\sqrt{y}} dy = 2 \int_{y=0}^1 (\sqrt{y} - y^2) dy \\ &= 2 \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{y^3}{3} \right)_0^1 = \frac{2}{3} \quad \dots\dots\dots(2)\end{aligned}$$

From (1) and (2), Green's theorem is verified.

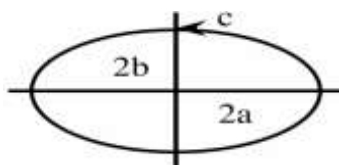
Problem 3. Show that area enclosed by a simple closed curve C is given by $\frac{1}{2} \oint_C \{x dy - y dx\}$. Using this, find the area bounded by the ellipse with axes $2a$ and $2b$.

Solution: we have

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$P = -y \Rightarrow \frac{\partial P}{\partial y} = -1 \text{ and } Q = x \Rightarrow \frac{\partial Q}{\partial x} = 1$$

$$\frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \iint_R (1+1) dx dy$$



To find the area of the ellipse, the equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

In parametric form

$$\begin{aligned}x &= a \cos \theta, & y &= b \sin \theta. \\ \Rightarrow dx &= -a \sin \theta, & dy &= b \cos \theta\end{aligned}$$

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \oint_C \{a \cos \theta b \cos \theta - b \sin \theta (-a \sin \theta)\} d\theta$$

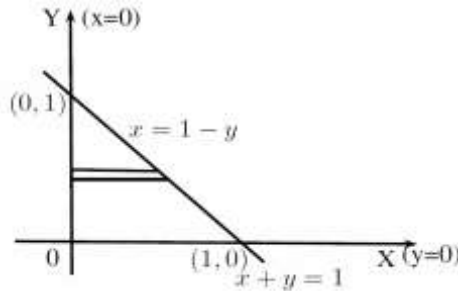
$$= \frac{1}{2} ab \int_0^{2\pi} (\cos^2 \theta + \sin^2 \theta) d\theta = \frac{1}{2} ab [2\pi] = \pi ab.$$

Problem 4. Using Green's theorem in the plane, evaluate $\int_C \{(2x^2 - y^2)dx + (x^2 + y^2)dy\}$, C is the boundary of the region bounded by $x = 0, y = 0, x + y = 1$.

Solution: We have

$$\oint_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$P = 2x^2 - y^2 \Rightarrow \frac{\partial P}{\partial y} = -2y \text{ and } Q = x^2 + y^2 \Rightarrow \frac{\partial Q}{\partial x} = 2x$$



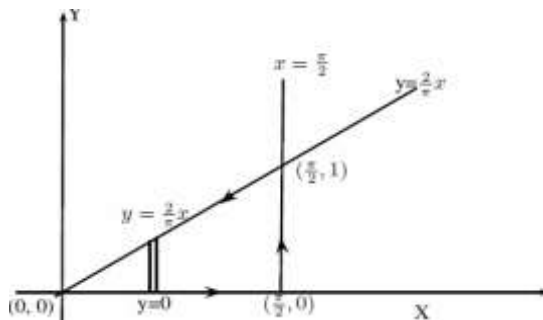
$$\begin{aligned} \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy &= \int_{y=0}^1 \int_{x=0}^{1-y} (2x + 2y) dx dy = 2 \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^{1-y} dy \\ &= 2 \int_0^1 \left(\frac{(1-y)^2}{2} + (1-y)y \right) dy \\ &= 2 \int_0^1 \left(\frac{1}{2}(1 + y^2 - 2y) + (y - y^2) \right) dy \\ &= 2 \left[\frac{1}{2} \left(y + \frac{y^3}{3} - 2 \frac{y^2}{2} \right) + \left(\frac{y^2}{2} - \frac{y^3}{3} \right) \right]_0^1 = \frac{2}{3}. \end{aligned}$$

Problem 5. Apply Green's theorem to evaluate $\int_C (y - \sin x)dx + \cos x dy$, where C is the triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$, and $y = \frac{2}{\pi}x$.

Solution: We have

$$\oint_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$P = y - \sin x \Rightarrow \frac{\partial P}{\partial y} = 1 \text{ and } Q = \cos x \Rightarrow \frac{\partial Q}{\partial x} = -\sin x$$



$$\int_C (y - \sin x)dx + \cos x dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\begin{aligned}
 &= \int_{x=0}^{\frac{\pi}{2}} \int_{y=0}^{\frac{2x}{\pi}} (-\sin x - 1) dy dx = - \int_0^{\frac{\pi}{2}} (\sin x + 1) [y]_0^{\frac{2x}{\pi}} dx \\
 &= - \frac{2}{\pi} \int_0^{\frac{\pi}{2}} (x \sin x + x) dx \\
 &= - \frac{2}{\pi} \left[-x \cos x + \sin x + \frac{x^2}{2} \right]_0^{\frac{\pi}{2}} = - \left(\frac{2}{\pi} + \frac{\pi}{4} \right).
 \end{aligned}$$

Exercise:

1. Verify Green's theorem for $\int_C (e^{-x} \sin y) dx + (e^{-x} \cos y) dy$, where C is the rectangle, whose vertices are $(0, 0)$, $(\pi, 0)$, $(\pi, \frac{\pi}{2})$ and $(0, \frac{\pi}{2})$. Ans: $[2(e^{-\pi} - 1)]$
2. Using Green's theorem, evaluate $\oint_C x^{-1} e^y dx + (e^y \ln x + 2x) dy$, where C is the bounded by $y = 2$, $y = x^4 + 1$. Ans: $\frac{16}{5}$
3. Using Green's theorem, evaluate $\oint_C (x^2 - \cosh y) dx + (y + \sin x) dy$ where C is the boundary of the rectangle $0 \leq x \leq \pi$, $0 \leq y \leq 1$. Ans: $\pi(\cosh 1 - 1)$

Surface Integral

Any integral which is to be evaluated over a surface is called surface integral.

Physical interpretation: The surface integral of a vector function $\vec{F}$ express the normal flux through a surface.

Surface and volume integrals help compute quantities distributed across surfaces or within three-dimensional regions. These concepts are widely applied in computer graphics, scientific computing, and 3D simulations, where properties like mass, heat, density, or energy must be calculated over objects or regions. They are also used in medical imaging, computational physics, and visualization systems for modelling and analysing physical phenomena.

Note: If $\vec{F}$ represents velocity vector of a fluid, the surface integral represents the rate of flow of fluid through the surface.

1. The surface integral of a vector point function $\vec{F}$ over a surface S is defined as the integral of normal component of $\vec{F}$ taken over the surface S .
2. If $\vec{F}$ represents the velocity of a fluid $\oiint_S \vec{F} \cdot \hat{n} ds$ gives the flux across the surface S .
3. If the flux of $\vec{F}$ across every closed surface S in a region R is zero. Then $\vec{F}$ is a solenoidal vector point function in the region R .
4. If $\vec{F}$ represents gravitational force, electric force or magnetic force in each case $\oiint_S \vec{F} \cdot \hat{n} ds$ gives corresponding flux.

Working Rule:

1. For the given surface ϕ , find $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$, $\hat{n}$ is outward unit normal vector to the surface.
2. Find $\vec{F} \cdot \hat{n}$
3. If the projection of S is taken in YZ -plane, then $ds = \frac{dy dz}{|\hat{n} \cdot \hat{i}|}$, where $\hat{i}$ is the unit vector along x - axis.
4. If the projection of S is taken in XY -plane, then $ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$, where $\hat{k}$ is the unit vector along z - axis.
5. If the projection of S is taken in XZ -plane, then $ds = \frac{dx dz}{|\hat{n} \cdot \hat{j}|}$, where $\hat{j}$ is the unit vector along y - axis.

NOTE: To evaluate any surface integral, it is convenient to evaluate the double integral of its projection on $xy, yz, \text{ or } zx$ plane.

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_S \vec{F} \cdot \vec{ds} = \iint_R \vec{F} \cdot \hat{n} \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$$

where R is the projection of S in XY - plane.

Problems 1. Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$ and S is the part of the $2x + 3y + 6z = 12$, located in first octant ($x = 0, y = 0, z = 0$).

Solution: Given $2x + 3y + 6z = 12 \Rightarrow \frac{x}{6} + \frac{y}{4} + \frac{z}{2} = 1$

Let $\phi = 2x + 3y + 6z - 12$

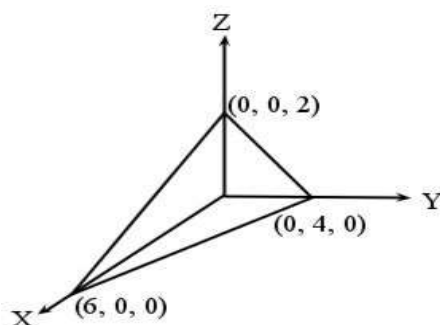
$$\text{then } \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{\sqrt{2^2 + 3^2 + 6^2}} = \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7}$$

$$\vec{F} \cdot \hat{n} = (18z\hat{i} - 12\hat{j} + 3y\hat{k}) \cdot \left(\frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7} \right) = \frac{36z - 36 + 18y}{7}$$

Projecting on to any plane (i. e, xy, yz or zx)

Projecting on to plane xy - plane



$$2x + 3y = 12; x: 0 \text{ to } 6; y: 0 \text{ to } \frac{12-2x}{3} \text{ and } |\hat{n} \cdot \hat{k}| = \frac{6}{7}$$

$$\begin{aligned}
 \iint_S \vec{F} \cdot \hat{n} \, ds &= \iint_R \vec{F} \cdot \hat{n} \frac{dx \, dy}{|\hat{n} \cdot \hat{k}|} = \int_{x=0}^6 \int_{y=0}^{\frac{12-2x}{3}} \frac{36z - 36 + 18y}{7} \frac{dy \, dx}{\frac{6}{7}} \\
 &= \frac{1}{6} \int_{x=0}^6 \int_{y=0}^{\frac{12-2x}{3}} \left(36 \frac{12-2x-3y}{6} - 36 + 18y \right) dy \, dx \\
 &= \frac{1}{6} \int_{x=0}^6 \int_{y=0}^{\frac{12-2x}{3}} (36 - 12x) dy \, dx = \frac{1}{6} \int_{x=0}^6 (36y - 12xy) \Big|_0^{\frac{12-2x}{3}} dx \\
 &= 2 \int_{x=0}^6 (3-x)y \Big|_0^{\frac{12-2x}{3}} dx = 2 \int_0^6 (3-x) \left(\frac{12-2x}{3} - 0 \right) dx \\
 &= 2 \int_0^6 (3-x) \frac{12-2x}{3} dx = 24.
 \end{aligned}$$

Problem 2. Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ where $\vec{F} = y\hat{i} + 2x\hat{j} - z\hat{k}$ and S is the surface of the plane $2x + y = 6$ included in the I octant cut by $z = 4$.

Solution: Let $\phi = 2x + y - 6$

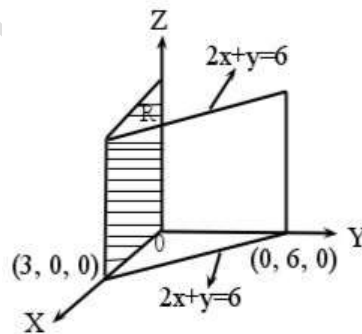
$$\text{then } \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k} = 2\hat{i} + \hat{j}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2\hat{i} + \hat{j}}{\sqrt{4+1}} = \frac{2\hat{i} + \hat{j}}{\sqrt{5}}$$

$$\vec{F} \cdot \hat{n} = (y\hat{i} + 2x\hat{j} - z\hat{k}) \cdot \left(\frac{2\hat{i} + \hat{j}}{\sqrt{5}} \right) = \frac{2y + 2x}{\sqrt{5}}$$

Projecting on to xz -plane, we get

$$|\hat{n} \cdot \hat{j}| = \frac{1}{\sqrt{5}}; \quad x: 0 \text{ to } 3 \text{ and } z: 0 \text{ to } 4$$



Now consider

$$\begin{aligned}
 \iint_S \vec{F} \cdot \hat{n} \, ds &= \iint_R \vec{F} \cdot \hat{n} \frac{dx \, dz}{|\hat{n} \cdot \hat{j}|} \\
 &= \int_{x=0}^3 \int_{z=0}^4 \frac{2y + 2x}{\sqrt{5}} \frac{1}{\frac{1}{\sqrt{5}}} dz \, dx = \int_{x=0}^3 \int_{z=0}^4 (2(6-2x) + 2x) dz \, dx
 \end{aligned}$$

$$= \int_{x=0}^3 \int_{z=0}^4 (12 - 2x) dz dx = \int_{x=0}^3 (12 - 2x) dx \int_{z=0}^4 1 dz = 108.$$

Problem 3. Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = z\hat{i} + x\hat{j} + 3y^2z\hat{k}$ where S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.

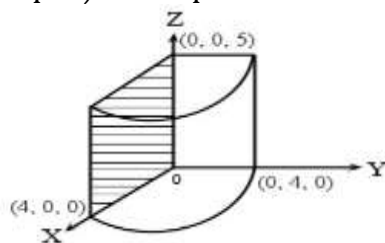
Solution: Given $x^2 + y^2 = 16$ is a right circular cylinder with base circle as $x^2 + y^2 = 16$, $z = 0$ and generates parallel to z -axis.

Let $\phi = x^2 + y^2 - 16$

$$\text{then } \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k} = 2x\hat{i} + 2y\hat{j}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{2(x\hat{i} + y\hat{j})}{2\sqrt{(x^2 + y^2)}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{16}} = \frac{x\hat{i} + y\hat{j}}{4}$$

$$\vec{F} \cdot \hat{n} = (z\hat{i} + x\hat{j} + 3y^2z\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{4}\right) = \frac{xz + xy}{4}$$



Projecting on to plane xz -plane

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} ds &= \iint_R \vec{F} \cdot \hat{n} \frac{dx dz}{|\hat{n} \cdot \hat{j}|} = \int_{z=0}^5 \int_{x=0}^4 \frac{xz + xy}{4} \frac{dx dz}{\frac{y}{4}} \quad \because |\hat{n} \cdot \hat{k}| = \frac{y}{4} \\ &= \int_{z=0}^5 \int_{x=0}^4 \left(\frac{xz}{y} + x\right) dx dz = \int_{z=0}^5 \int_{x=0}^4 \left(\frac{xz}{\sqrt{16-x^2}} + x\right) dx dz \\ &= \int_{z=0}^5 z dz \int_{x=0}^4 \frac{x}{\sqrt{16-x^2}} dx + \int_{x=0}^4 x dx \int_{z=0}^5 1 dz = 90. \end{aligned}$$

Exercise:

- Find the surface integral over the parallelepiped $x = 0, y = 0, x = 1, y = 2, z = 3$ when $\vec{A} = 2xy\hat{i} + yz^2\hat{j} + xz\hat{k}$ Ans: 33.
- If S is the surface of the sphere $x^2 + y^2 + z^2 = d^2$ and $\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k}$, evaluate $\iint_S \vec{A} \cdot \hat{n} ds$. Ans: $\frac{2\pi d^3}{3}(a + b + c)$
- If $\vec{F} = 2y\hat{i} - 3\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, show that $\iint_S \vec{F} \cdot \hat{n} ds = 132$.

Gauss divergence theorem

(Relation between surface and volume integrals)

Statement: If V is the volume bounded by a closed surface S and $\vec{F}$ is a vector point function having continuous derivatives, then

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV,$$

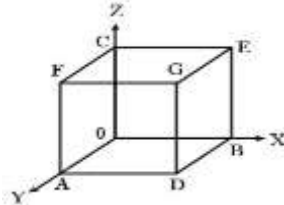
where $\hat{n}$ is the unit normal drawn to S . ($\hat{n} \rightarrow$ outward unit normal i.e, normal vector away from the surface)

The Gauss divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field inside the enclosed volume. In computer science, it is useful in computational physics, fluid dynamics simulations, and numerical modelling, where it simplifies calculations by converting volume integrals into surface integrals in large-scale simulations.

Problem 1. Verify Gauss divergence theorem for $\vec{F} = (x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}$ taken over the surface of the cube bounded by the planes $x = y = z = 2$ and the coordinate planes.

Solution: We have to verify that

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$



$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} \, ds &= \int_{S_1} \vec{F} \cdot \hat{n} \, ds + \int_{S_2} \vec{F} \cdot \hat{n} \, ds + \int_{S_3} \vec{F} \cdot \hat{n} \, ds + \int_{S_4} \vec{F} \cdot \hat{n} \, ds + \int_{S_5} \vec{F} \cdot \hat{n} \, ds \\ &\quad + \int_{S_6} \vec{F} \cdot \hat{n} \, ds \end{aligned}$$

where $S_1, S_2, S_3, S_4, S_5, S_6$ are the six faces of the cube.

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_S ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds$$

For S_1 (DBEG) which is parallel to yz -plane its equation is $x = 2, \hat{n} = \hat{i}$ & $ds = dydz$.

Here $\hat{n} = \hat{i}$, $(x^3 - yz)\hat{i} \cdot \hat{i} = x^3 - yz$ (remaining are zero).

$$\begin{aligned} \iint_{S_1} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds &= \iint_{S_1} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{i} \, dy \, dz \\ &= \int_{z=0}^2 \int_{y=0}^2 (8 - yz) \, dy \, dz = \int_{z=0}^2 \left[8y - \frac{zy^2}{2} \right]_0^2 \, dz = \int_{z=0}^2 [16 - 2z] \, dz \\ &= \left[16z - \frac{2z^2}{2} \right]_0^2 = 32 - 4 = 28. \end{aligned}$$

For S_2 (OCEB) which is xz -plane, $y = 0$, $\hat{n} = -\hat{j}$ & $ds = dz \, dx$.

$$\iint_{S_2} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds = \int_{x=0}^2 \int_{y=0}^2 (2x^2y) \, dz \, dx = 0 \quad \because y = 0$$

For S_3 ($OADB$) which is xy - plane, $z = 0$, $\hat{n} = -\hat{k}$ & $ds = dx \, dy$.

$$\iint_{S_3} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds = \int_{y=0}^2 \int_{x=0}^2 (-z) \, dx \, dy = 0 \quad \because z = 0$$

For S_4 ($OCFA$) which is yz - plane, $x = 0$, $\hat{n} = -\hat{i}$ & $ds = dy \, dz$.

$$\begin{aligned} \iint_{S_4} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds \\ = \int_{z=0}^2 \int_{y=0}^2 -(x^3 - yz) \, dy \, dz = \int_{z=0}^2 \int_{y=0}^2 yz \, dy \, dz = \int_{z=0}^2 z \, dz \int_{y=0}^2 y \, dy \\ = \left[\frac{z^2}{2} \right]_0^2 \left[\frac{y^2}{2} \right]_0^2 = 4. \end{aligned}$$

For S_5 ($GFAD$) which is parallel to xz - plane, its equation is $y = 2$, $\hat{n} = \hat{j}$ & $ds = dx \, dz$.

$$\begin{aligned} \iint_{S_5} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds \\ = \int_{z=0}^2 \int_{x=0}^2 -2x^2y \, dx \, dz = \int_{z=0}^2 \int_{x=0}^2 -2x^2(2) \, dx \, dz = -4 \int_{x=0}^2 x^2 \, dx \int_{z=0}^2 1 \, dz = \\ = -4 \left[\frac{x^3}{3} \right]_0^2 [z]_0^2 = -4 \times \frac{8}{3} \times 2 = -\frac{64}{3}. \end{aligned}$$

For S_6 ($GE CF$) which is parallel to xy - plane, its equation is $z = 2$, $\hat{n} = \hat{k}$ & $ds = dx \, dy$.

$$\begin{aligned} \iint_{S_6} ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds \\ = \int_{x=0}^2 \int_{y=0}^2 z \, dy \, dx = 2 \int_{x=0}^2 1 \, dx \int_{y=0}^2 1 \, dy = 8. \end{aligned}$$

$$\therefore \iint_S ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) \cdot \hat{n} \, ds = 28 + 0 + 0 + 4 - \frac{64}{3} + 8 = \frac{56}{3}.$$

Now to evaluate $\iiint_V \nabla \cdot \vec{F} \, dV$

Consider

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot ((x^3 - yz)\hat{i} - 2x^2y\hat{j} + z\hat{k}) = 3x^2 - 2x^2 + 1 = x^2 + 1$$

$$\begin{aligned} \therefore \iiint_V \nabla \cdot \vec{F} dV &= \int_{z=0}^2 \int_{y=0}^2 \int_{x=0}^2 (x^2 + 1) dx dy dz = \int_{z=0}^2 1 dz \int_{y=0}^2 1 dy \int_{x=0}^2 (x^2 + 1) dx \\ &= [z]_0^2 [y]_0^2 \left[\frac{x^3}{3} + x \right]_0^2 = (2)(2) \left(\frac{8}{3} + 2 \right) = \frac{56}{3} \quad \dots \dots \dots (2) \end{aligned}$$

From (1) and (2), we see that

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dV = \frac{56}{3}$$

Hence the Gauss divergence theorem.

Problem 2. Verify divergence theorem $\vec{F} = xy\hat{i} - y\hat{j} + 2z\hat{k}$ over the region bounded by the plane $x = 0, y = 0, z = 0$ & $2x + 2y + z = 4$.

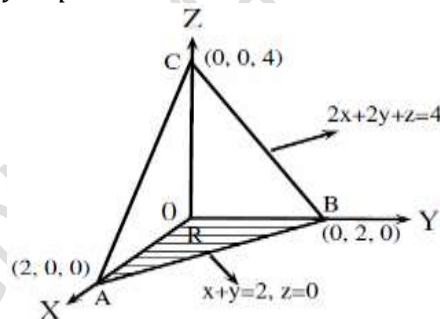
Solution: Let $\phi = 2x + 2y + z - 4$

$$i.e. \quad \frac{x}{2} + \frac{y}{2} + \frac{z}{4} = 1$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2\hat{i} + 2\hat{j} + \hat{k}}{3}$$

$$\vec{F} \cdot \hat{n} = \frac{2xy - 2y + 2z}{3}$$

Now project the surface on xy - plane



$$|\hat{n} \cdot \hat{k}| = \left| \frac{2\hat{i} + 2\hat{j} + \hat{k}}{3} \cdot \hat{k} \right| = \frac{1}{3}$$

$x: 0 \text{ to } 2$

$y: 0 \text{ to } 2 - x$

plane $2x + 2y + z = 4 \Rightarrow z = 4 - 2x - 2y$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iint_R \vec{F} \cdot \hat{n} \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \iint_R \left(\frac{2xy - 2y + 2z}{3} \right) \frac{dx dy}{\frac{1}{3}}$$

$$\begin{aligned}
 &= 2 \int_{x=0}^2 \int_{y=0}^{2-x} (xy - y + 4 - 2x - 2y) dy dx \\
 &= 2 \int_{x=0}^2 \int_{y=0}^{2-x} (xy - 3y - 2x - 2y + 4) dy dx \\
 &= 2 \int_{x=0}^2 \left[\frac{xy^2}{2} - \frac{3y^2}{2} - 2xy + 4y \right]_0^{2-x} dx \\
 &= 2 \int_{x=0}^2 \left[\frac{x(2-x)^2}{2} - \frac{3(2-x)^2}{2} - 2x(2-x) + 4(2-x) \right] dx \\
 &= 2 \int_{x=0}^2 \left[\frac{1}{2}(4x + x^3 - 4x^2) - \frac{3}{2}(4 + x^2 - 4x) - 8x + 2x^2 + 8 \right] dx \\
 &= 2 \left[\frac{1}{2} \left(\frac{4x^2}{2} + \frac{x^4}{4} - \frac{4x^3}{3} \right) - \frac{3}{2} \left(4x + \frac{x^3}{3} - \frac{4x^2}{2} \right) - \frac{8x^2}{2} + \frac{2x^3}{3} + 8x \right]_0^2 \\
 &= 4 \quad (1)
 \end{aligned}$$

Surface	Remarks	$\hat{n}$	ds	$\vec{F} \cdot \hat{n}$
$S_1: AOB$	$xy - plane$ $z = 0$	$\hat{n} = -\hat{k}$	$dx dy$	$-2z = 0$
$S_2: BOC$	$yz - plane$ $x = 0$	$\hat{n} = -\hat{i}$	$dy dz$	$-xy = 0$
$S_3: AOC$	$xz - plane$ $y = 0$	$\hat{n} = -\hat{j}$	$dx dz$	$y = 0$
$S_4: ABC$	Projection on $xy - plane$	$\hat{n} = \frac{\nabla \phi}{ \nabla \phi }$ $= \frac{2\hat{i} + 2\hat{j} + \hat{k}}{3}$	$\frac{dx dy}{ \hat{n} \cdot \hat{k} }$	$\vec{F} \cdot \hat{n}$ $= \frac{2(xy - y + 4 - 2x - 2y)}{3}$

Now consider

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xy\hat{i} - y\hat{j} + 2z\hat{k}) = y + 1$$

$$\begin{aligned}
 \therefore \iiint_V \nabla \cdot \vec{F} dV &= \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} (y+1) dz dy dx \\
 &= \int_{x=0}^2 \int_{y=0}^{2-x} [(y+1)z]_{z=0}^{4-2x-2y} dy dx \\
 &= \int_{x=0}^2 \int_{y=0}^{2-x} [(y+1)(4-2x-2y)] dy dx \\
 &= \int_{x=0}^2 \int_{y=0}^{2-x} [2y + 4 - 2xy - 2x - 2y^2] dy dx \\
 &= \int_0^2 \left[\frac{2y^2}{2} + 4y - \frac{2xy^2}{2} - 2xy - \frac{2y^3}{3} \right]_0^{2-x} dy
 \end{aligned}$$

$$= \int_0^2 \left[(2-x)^2 + 4(2-x) - x(2-x)^2 - 2x(2-x) - \frac{2}{3}(2-x)^3 \right] dy$$

$$= 4 \dots (2)$$

From (1) and (2), Gauss divergence theorem is verified.

Problem 3. Using divergence theorem, evaluate $\iint_S [(x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}] \cdot \hat{n} \, ds$, over the surface of the rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.

Solution: We have

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot ((x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}) = 2(x + y + z)$$

$$\therefore \iiint_V \nabla \cdot \vec{F} \, dV = \int_{z=0}^c \int_{y=0}^b \int_{x=0}^a 2(x + y + z) \, dx \, dy \, dz$$

$$= 2 \int_{z=0}^c \int_{y=0}^b \left(\frac{x^2}{2} + yx + zx \right)_0^a \, dy \, dz$$

$$= 2 \int_{z=0}^c \int_{y=0}^b \left(\frac{a^2}{2} + ay + az \right) \, dy \, dz$$

$$= 2 \int_0^c \left[\frac{a^2}{2} y + \frac{ay^2}{2} + azy \right]_0^b \, dz = 2 \int_0^c \left[\frac{a^2 b}{2} + \frac{ab^2}{2} + azb \right] \, dz$$

$$= \left[\frac{a^2 b}{2} z + \frac{ab^2}{2} z + ab \frac{z^2}{2} \right]_0^c = abc(a + b + c).$$

Problem 4. Evaluate using divergence theorem $\iint_S [x^3\hat{i} + x^2y\hat{j} + x^2z\hat{k}] \cdot \hat{n} \, ds$, where S is the surface consisting of the cylinder $x^2 + y^2 = a^2$ and the circular discs cut by the plane $z = 0$ & $z = b$.

Solution: Here

$$\vec{F} = x^3\hat{i} + x^2y\hat{j} + x^2z\hat{k}$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^3\hat{i} + x^2y\hat{j} + x^2z\hat{k}) = 5x^2$$

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV = \iiint_V 5x^2 \, dV$$

using cylindrical coordinates

$$x = r \cos \theta,$$

$$y = r \sin \theta,$$

$$z = z,$$

$$dV = dx \, dy \, dz = r \, dr \, d\theta \, dz$$

$$r: 0 \text{ to } a$$

$$\theta: 0 \text{ to } 2\pi$$

$z: 0 \text{ to } b$

$$\begin{aligned}\therefore \iiint_V 5x^2 dV &= \int_{z=0}^b \int_{\theta=0}^{2\pi} \int_{r=0}^a 5(r^2 \cos \theta) r dr d\theta dz \\ &= 5 \int_{r=0}^a r^3 dr \int_{\theta=0}^{2\pi} \cos^2 \theta d\theta \int_{z=0}^b 1 dz = 5 \frac{a^4}{4} \times \frac{1}{2} \times \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi} \times [z]_0^b \\ &= \frac{5a^4 b \pi}{4}.\end{aligned}$$

Problem 5. Using divergence theorem, $\iint_S \vec{F} \cdot \hat{n} ds$, $\vec{F} = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$ taken over the surface consisting of the hemisphere $x^2 + y^2 + z^2 = a^2$ above the xy -plane bounded by the xy -plane.

Solution: Here

$$\vec{F} = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}) = 3(x^2 + y^2 + z^2) = 3a^2$$

Using Spherical coordinates

$$\because x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

$r: 0 \text{ to } a$

$\theta: 0 \text{ to } \frac{\pi}{2}$ [vertical angle]

$\phi: 0 \text{ to } 2\pi$ [Horizontal angle]

$$\begin{aligned}\iint_S \vec{F} \cdot \hat{n} ds &= \iiint_V \text{div}(\vec{F}) dV = 3 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\frac{\pi}{2}} \int_{r=0}^a r^2 (r^2 \sin \theta) dr d\theta d\phi \\ &= 3 \int_{\phi=0}^{2\pi} 1 d\phi \times \int_{\theta=0}^{\frac{\pi}{2}} \sin \theta d\theta \times \int_{r=0}^a r^4 dr \\ &= 3 \times 2\pi \times \frac{a^5}{5} \times 1 = \frac{6\pi a^5}{5}.\end{aligned}$$

Using Cartesian coordinates

$$\begin{aligned}\iint_S \vec{F} \cdot \hat{n} ds &= \iiint_V \text{div}(\vec{F}) dV \\ &= \int_{x=-a}^a \int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_{z=0}^{\sqrt{a^2-x^2-y^2}} 3(x^2 + y^2 + z^2) dz dy dx = \frac{6\pi a^5}{5}\end{aligned}$$

Exercise:

1. Verify divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.
2. Using divergence theorem, evaluate $\iint_S \vec{r} \cdot \hat{n} ds$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 9$. Ans: 108π

3. Using divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ over the entire surface S of the region above xy plane bounded by the cone $x^2 + y^2 = z^2$ the plane $z = 4$ where $\vec{F} = 4xz\hat{i} - xyz^2\hat{j} + 3z\hat{k}$ Ans: 704π

STOKES THEOREM

(Relation between line and surface integral)

Statement: If S be an open surface bounded by a simple closed curve C and $\vec{F}$ be any vector point function having continuous first order partial derivatives, then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \iint_S \text{curl } \vec{F} \cdot d\vec{s}$$

where $\hat{n}$ is the outward drawn unit normal at any point to S .

Stokes' theorem relates the circulation of a vector field around a closed curve to the surface integral of the curl over the surface bounded by the curve. In computer science, it is applied in fluid flow simulations, electromagnetic field modelling, and physics-based animation, where rotational properties of vector fields need to be analysed in simulations and graphical environments.

Problem 1. Verify Stokes theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the rectangle bounded by the lines $x = \pm a, y = 0, y = b$.

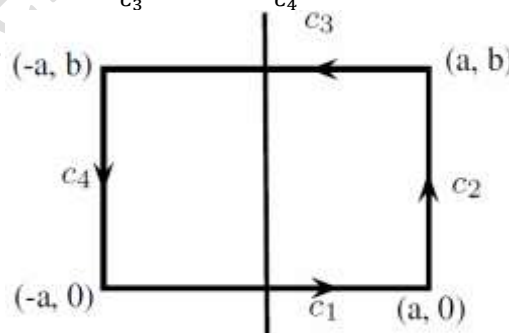
Solution: We have to prove that

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Now

$$\vec{F} \cdot d\vec{r} = ((x^2 + y^2)\hat{i} - 2xy\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = (x^2 + y^2)dx - 2xy \, dy$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} + \int_{C_3} \vec{F} \cdot d\vec{r} + \int_{C_4} \vec{F} \cdot d\vec{r} \quad \dots \dots \dots (1)$$



$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C (x^2 + y^2)dx - 2xy \, dy$$

Along $C_1: y = 0 \Rightarrow dy = 0, x: -a \text{ to } a$

$$\therefore \int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_1} (x^2 + y^2)dx - 2xy \, dy = \int_{-a}^a x^2 dx = \frac{2a^3}{3}$$

Along $C_2: x = a \Rightarrow dx = 0$, $y: 0$ to b

$$\therefore \int_{C_2} \vec{F} \cdot d\vec{r} = \int_{C_2} (x^2 + y^2)dx - 2xy dy = \int_0^b 2ay dy = -ab^2$$

Along $C_3: y = b \Rightarrow dy = 0$, $x: a$ to $-a$

$$\therefore \int_{C_3} \vec{F} \cdot d\vec{r} = \int_{C_3} (x^2 + y^2)dx - 2xy dy = \int_a^{-a} (x^2 + b^2) dx = -\frac{2a^3}{3} - 2ab^2$$

Along $C_4: x = -a \Rightarrow dx = 0$, $y: b$ to 0

$$\therefore \int_{C_4} \vec{F} \cdot d\vec{r} = \int_{C_4} (x^2 + y^2)dx - 2xy dy = \int_b^0 -2(-a)y dy = -ab^2$$

$$\therefore (1) \Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2 \quad \dots \dots \dots (2)$$

Next, consider

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} = -4y\hat{k}$$

Rectangle in xy - plane $\Rightarrow \hat{n} = \hat{k}$ and $ds = dx dy$

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} ds = \iint_R -4y\hat{k} \cdot \hat{k} dx dy = - \int_{y=0}^b \int_{x=-a}^a 4y dx dy = -4 \int_{-a}^a 1 dx \times \int_0^b y dy$$

$$\therefore \iint_S \text{curl } \vec{F} \cdot \hat{n} ds = -4ab^2 \quad \dots \dots \dots (3)$$

From (2) and (3), Stokes theorem is verified.

Problem 2. Verify Stokes theorem for $\vec{F} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$ bounded by its projection on xy - plane.

Solution: The projection of upper half of the sphere $x^2 + y^2 + z^2 = 1$ in the xy - plane ($z = 0$) is the circle $x^2 + y^2 = 1$ and let C be its boundary.

We have

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds$$

Consider

$$\oint_C \vec{F} \cdot d\vec{r} = \int_C \{(2x - y)dx - yz^2 dy - y^2z dz\}$$

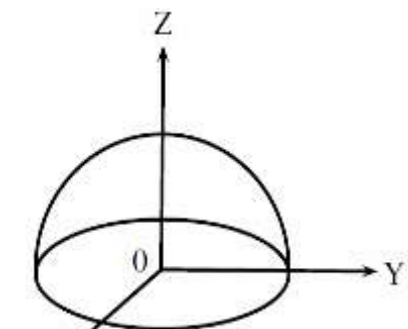
In xy - plane, $z = 0 \Rightarrow dz = 0$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \int_C \{(2x - y)dx - yz^2 dy - y^2z dz\} = \int_C (2x - y)dx$$

Here C is the circle $x^2 + y^2 = 1$ whose parametric equation is given by

$$\begin{aligned} x &= \cos \theta, & y &= \sin \theta \\ \Rightarrow dx &= -\sin \theta d\theta, & dy &= \cos \theta d\theta \end{aligned}$$

Here $\theta: 0$ to 2π .



$$\begin{aligned}\therefore \oint_C \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} (2 \cos \theta - \sin \theta)(-\sin \theta) d\theta = \int_0^{2\pi} (\sin 2\theta + \sin^2 \theta) d\theta = 0 + \int_0^{2\pi} \sin^2 \theta d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta = 4 \times \frac{1}{2} \times \frac{\pi}{2} = \pi.\end{aligned}$$

Next, consider

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix} = \hat{k}$$

On the xy -plane $\hat{n} = \hat{k}$ and $ds = dx dy$

$$\begin{aligned}\iint_S \text{curl } \vec{F} \cdot \hat{n} ds &= \iint_R \hat{k} \cdot \hat{k} dx dy = \iint_R 1 dx dy \\ &= \text{Area of circle } (x^2 + y^2 = 1) = \pi \quad \because r = 1\end{aligned}$$

Hence Stokes theorem is verified.

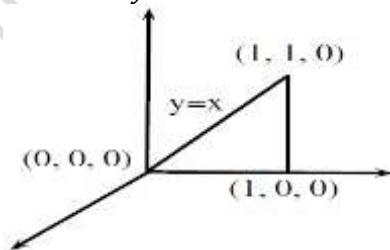
Problem 3. Evaluate by Stokes theorem $\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$, C is the boundary of the triangular with vertices $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.

Solution: By Stokes theorem we have

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} ds$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix} = 2\hat{i} + \hat{k}$$

In xy -plane $\Rightarrow \hat{n} = \hat{k}$ and $ds = dx dy$



$$\begin{aligned}\iint_S \text{curl } \vec{F} \cdot \hat{n} ds &= \iint_R (2\hat{i} + \hat{k}) \cdot \hat{k} dx dy = \iint_R 1 dx dy = \text{Area of the triangle} \\ &= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}.\end{aligned}$$

Exercise:

1. Evaluate $\oint_C xy dx + xy^2 dy$ by Stoke's theorem where C is the square in the xy plane with vertices $(1,0)$ $(-1,0)$ $(0,1)$ $(0,-1)$.
2. Verify Stokes's theorem where $\vec{A} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ and S : upper half of the surface of the sphere $x^2 + y^2 + z^2 = 1$ Ans: π



3. Evaluate $\oint_C 4z \, dx - 2x \, dy + 2x \, dz$ by Stoke's theorem where C is the ellipse $x^2 + y^2 = 1$, $z = y + 1$. Ans: -4π

Video Links:

1. Line integral
<https://www.youtube.com/watch?v=7FUNdFN6ZKI>
2. Surface integral
<https://www.youtube.com/watch?v=I1dfwKPV75A>

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Department of Mathematics, RVCE